
CivicPulse: Smart Civic Complaint & Issue Management System

Hackathon Case Study Submission – Stage 2

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Product Overview

CivicPulse centralizes municipal reporting by connecting citizens and city operators through an AI-powered, multi-channel platform.



Multi-Channel Access

Citizens report issues smoothly via a mobile-first PWA or accessible voice helpline.



AI-Assisted Workflow

AI automatically categorizes, summarizes, and structures complaints for faster routing.



Location Intelligence

Precise mapping surfaces existing issues to citizens and groups recurring problems for admins.



Structured Tracking

Every report moves through strict stages: New → Assigned → In Progress → Resolved.

The Problem

Current civic systems suffer from a severe disconnect. Citizens struggle with complex interfaces, while municipalities drown in unstructured data.

For Citizens

- ⚠️ Unclear department routing and complex digital forms deter engagement.
- ⚠️ Total lack of visibility into existing reports or resolution progress.
- ⚠️ Strictly digital platforms exclude users without mobile data access.

For Municipal Teams

- ⚠️ Overwhelmed by massive volumes of unstructured, uncategorized complaints.
- ⚠️ Inability to quickly identify high-priority emergencies amidst routine tasks.
- ⚠️ Lack of clear ownership tracking and automated citizen communication.

Citizen Experience

The platform guides users through a transparent journey, prioritizing community awareness and seamless tracking.

01

Streamlined Reporting

Quickly enter category, description, photos, and precise GPS location.

02

Contextual Discovery

System surfaces similar, nearby complaints proactively before creating duplicates.

03

Citizen Agency

Users choose to upvote an existing report or submit a new one—no forced automated merges.

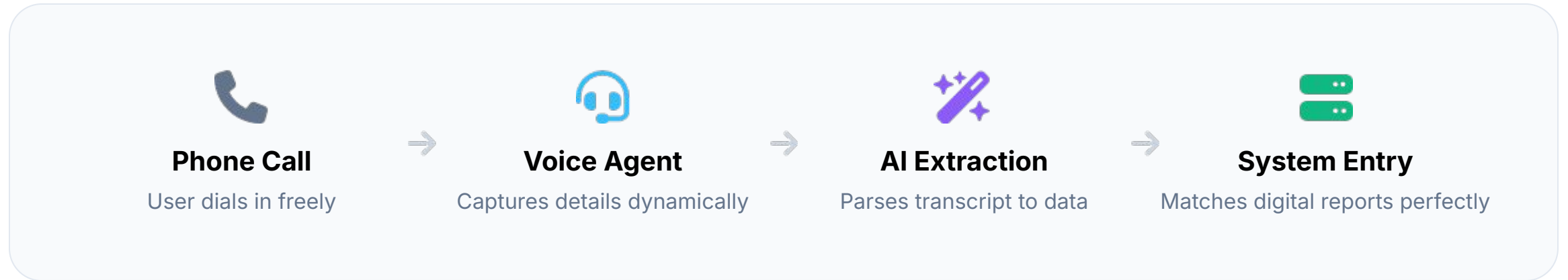
04

Continuous Tracking

Live timeline views, priority updates, and a regional feed of neighborhood resolutions.

Voice Helpline Intake

Our automated voice system bridges the digital divide, ensuring equitable reporting access for all demographics without requiring data plans.



The AI instantly extracts category, location, summary, and urgency from the natural conversation, creating a standardized work item for municipal teams.

AI Components & Boundaries

Artificial Intelligence acts purely as an operational accelerator. It processes unstructured data while strictly leaving final authority to human administrators.

Classification

Converts lengthy natural-language text into precise municipal operational categories.

Summarization

Condenses long, rambling descriptions or call transcripts into crisp operational summaries.

Data Extraction

Pulls structured data points like addresses, cross-streets, and landmarks from free text.

Discovery

Semantically ranks potentially relevant existing reports near the newly submitted location.



System Boundary: AI does NOT control user authentication, status transitions, final operational priorities, or core audit records.

Explainable Prioritization

We replace opaque "black box" models with a transparent, rule-based scoring engine that balances citizen urgency with objective operational metrics.



Citizen-reported urgency is an input, but the final score relies on robust data logic.

Transparent Score Breakdown

Base Issue Category (e.g. Water Main Break) **+27 pts**

Complaint Age (SLA Escalation) **+24 pts**

Community Upvotes (Engagement) **+15 pts**

Geographic Density (Nearby similar issues) **+12 pts**

Location Intelligence

Precise geospatial mapping transforms isolated complaints into actionable regional heatmaps, improving routing and response times.

Precise Anchoring

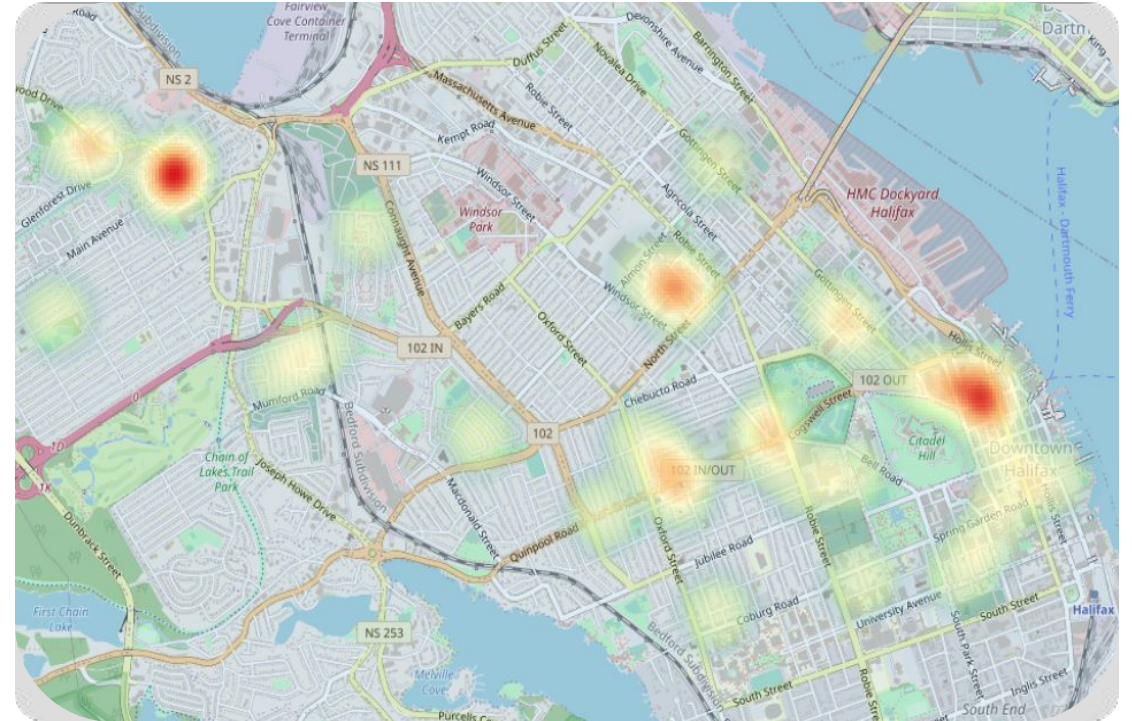
Every complaint is saved with exact latitude/longitude coordinates.

Contextual Boundaries

Citizen-facing discovery is deliberately restricted to a 50 km radius to maintain local relevance.

Operational Mapping

Enables spatial clustering, heatmaps, and dynamic dispatch routing for municipal teams.



Municipal Operations Dashboard

A centralized command center equipping city operators to efficiently monitor workloads, triage incoming data, and track resolutions.



Global Oversight

Real-time view of open, assigned, and resolved queues alongside SLA aging warnings.



Complaint Management

Operators review AI summaries, correct routing tags, and add internal or public status notes.



Performance Analytics

Tracks departmental resolution times, channel intake volumes (Voice vs Web), and category trends.

Technical Architecture

Designed with a decoupled, API-first structure to ensure rapid deployment, high availability, and seamless third-party integration.



Security & Privacy

We strictly enforce data minimization protocols and granular access controls to protect citizen identities and ensure audit integrity.



Identity Protection

Phone numbers are never public. Fully anonymous reporting is natively supported.



Access Boundaries

Strict role-based limits for citizens vs operators. Internal admin notes remain private.



Audit Integrity

Every administrative status change is logged into an immutable organizational audit trail.



Platform Defense

Protected continuously by strict API payload validation, file-upload checks, and rate limits.

System Reliability

Designed for graceful degradation, the architecture guarantees core functionality remains active even during external service outages.



AI Independence

If AI categorization fails, reports are safely saved and routed for manual human review.



Redundant Channels

If voice telephony goes offline, the robust web PWA remains fully operational.



Isolated Providers

Geocoding, AI, and SMS APIs are isolated modules that can be swapped without core rewrites.

Research & Foundations

Our methodology is anchored in proven successes from the world's largest civic tech deployments, modernized with AI efficiency.

NYC311 Scale

Validates the extreme necessity of multi-channel tracking, supporting over 36.5 million contacts annually.

CPGRAMS Metrics

Demonstrates the operational requirement for large-scale disposal tracking and tight departmental accountability.

SeeClickFix Visibility

Highlights the massive community value generated by transparent, map-based public issue discovery.

CivicPulse Evolution

Enhances these legacy models with instant AI structuring, explainable triage scores, and voice inclusivity.

Expected Impact

CivicPulse dramatically reduces municipal resolution timelines while simultaneously fostering deeper community engagement and public trust.

For Citizens

- ✓ Lowered reporting barriers via automated, conversational voice systems.
- ✓ Transparent visibility into localized neighborhood issues before reporting.
- ✓ Active participation fostered through timeline tracking and community upvoting.

For Municipal Teams

- ✓ Standardized, structured data extracted instantly from all intake channels.
- ✓ Faster workflow routing backed by explainable, rule-based triage logic.
- ✓ Data-driven geographic visibility for targeting recurring infrastructure gaps.

Product Differentiation

Unlike generic 311 forms, CivicPulse proactively structures incoming data to eliminate tedious manual operator triage entirely.

Dual Access Reality

Voice channels are not just chatbots; they generate identical, structured municipal work items matching the PWA.

Automated Structuring

AI comprehensively removes manual reading, classification, and summarization from the operator workload.

Deliberate Citizen Control

Users choose to support complaints manually, sidestepping the disastrous data-loss pitfalls of automated merges.

Radical Transparency

Delivers fully explainable triage scoring criteria rather than relying on confusing, biased neural-network ranking.



Thank You
